DATA ETHICS

Complete Detailed Study Notes • B.Sc. Computer Science with Artificial Intelligence

Five units | Theory explanations | Tables | Examples | Mind maps | Exam revision

These notes follow the syllabus provided. The explanations use student-friendly English and are structured for 5-mark and 10-mark semester answers. Legal frameworks can evolve; the notes distinguish the withdrawn Indian PDP Bill from the enacted DPDP Act, 2023.

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• Unit II – Privacy Laws and Regulatory Frameworks

• Unit III – Ethical Challenges in AI

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• Quick revision tables and important 10-mark questions

UNIT I – INTRODUCTION TO DATA ETHICS

1. Ethics in Data Science: Principles and Challenges

Data Ethics is the branch of ethics that deals with moral principles governing the collection, storage, processing, analysis, sharing, and use of data. It ensures data is handled responsibly, fairly, transparently, and securely. Data science can reveal useful patterns, but datasets may include personal or sensitive information; ethical practices protect individuals and society.

Importance: Data ethics protects privacy, promotes fairness, builds trust, ensures transparency, prevents misuse, supports accountability, and improves responsible decision-making.

Principle	Explanation	Example
Privacy	Protect personal and sensitive information	Keep student marks confidential
Transparency	Explain data collection and use	Publish a clear privacy notice
Fairness	Avoid unjust discrimination	Evaluate applicants using fair criteria
Accountability	Take responsibility for data-related actions	Report and address a data leak
Consent	Obtain permission when consent is required	Ask before accessing location
Security	Protect against unauthorized access	Use encryption and access controls
Accuracy	Maintain correct and reliable data	Update incorrect customer details
Purpose limitation	Use data for specified purposes	Use attendance data for attendance
Data minimization	Collect only necessary information	Avoid unnecessary personal fields

Challenges in Data Ethics

• **Privacy violations:** data may be collected or used beyond what people expect.

• **Algorithmic bias:** biased data can lead to unfair predictions.

• **Transparency:** complex systems may be difficult to explain.

• **Security:** datasets may be exposed through hacking or unauthorized access.

• **Ownership and control:** individuals may not know who controls their data.

• **Consent:** users may not understand lengthy terms and conditions.

• **Profiling:** detailed profiles can be created and used in ways that harm people.

Example: An online retailer collects names, addresses, and purchase histories. Sharing these records with other companies without a proper legal basis or meaningful notice creates ethical concerns.

2. Ethical Theories: Utilitarianism, Deontology, and Virtue Ethics

Ethical theories provide structured ways to judge whether an action is morally right. They guide data professionals when a decision involves competing values.

A. Utilitarianism

Utilitarianism states that an action is morally right when it produces the greatest overall benefit or happiness for the greatest number. It focuses on consequences, weighing benefits and harms for all affected people.

• Promotes overall welfare and seeks to reduce harm.

• Considers outcomes when choosing between actions.

• Example: an AI traffic system adjusts signals to reduce congestion for many commuters.

• **Limitation:** the welfare or rights of a minority may be overlooked, and consequences can be hard to measure.

B. Deontology

Deontology is an ethical theory that emphasizes moral duties, rules, and principles regardless of the final outcome. It is associated with Immanuel Kant. It asks whether an action follows a moral duty and respects people as individuals.

• Respect rights, privacy, honesty, and consent.

• Do not deceive or misuse confidential information.

• Example: a company refuses to disclose a customer's confidential data even if disclosure would earn profit.

• **Limitation:** strict rules may be difficult when duties conflict.

C. Virtue Ethics

Virtue Ethics focuses on a person's moral character and qualities rather than only consequences or rules. It is associated with Aristotle. Important virtues include honesty, integrity, fairness, responsibility, courage, and compassion.

• Encourages professionals to develop good character.

• Supports trustworthy and responsible conduct.

• Example: a data scientist openly reports a model's bias and works to fix it.

• **Limitation:** people may disagree about which virtues matter most in a situation.

Theory	Main focus	Question to remember	Data example
Utilitarianism	Consequences	What creates the greatest overall good?	Traffic AI reduces delays
Deontology	Duty and rules	What is my moral duty?	Protect confidential data
Virtue Ethics	Character	What would an honest, fair person do?	Report model errors

3. Responsible AI and the Role of Ethics in Automation

Responsible AI means designing, developing, deploying, and using AI in ways that are fair, safe, transparent, accountable, privacy-respecting, and aligned with human values. Responsible systems should not cause unnecessary harm or unjust discrimination.

Principle	Theory content
Fairness	AI should avoid unjust discrimination and consider impacts across groups.
Transparency	Users should receive understandable information about important AI decisions.
Accountability	Organizations and people responsible for AI should answer for its outcomes.
Privacy	Personal data should be protected and processed appropriately.
Safety and reliability	Systems should be tested and work consistently without unacceptable harm.
Human oversight	People should be able to supervise, review, or intervene in important decisions.
Inclusiveness	Systems should be designed to work for a broad range of users.

Automation is the use of machines or software to perform tasks with reduced human intervention. Ethics in automation prevents discrimination, protects workers, preserves human control, promotes safety, clarifies responsibility, and limits unnecessary data collection. Example: an AI admission tool may assist in reviewing applications, but meaningful human review can help detect unfair rejection.

4. Ethical Issues in Big Data and Cloud Computing

Big Data refers to very large, complex, and rapidly generated datasets that require advanced methods to store, process, and analyze. It is used in healthcare, banking, education, transport, and business.

The 5 Vs	Meaning
Volume	Large quantity of data
Velocity	Speed at which data is generated and processed

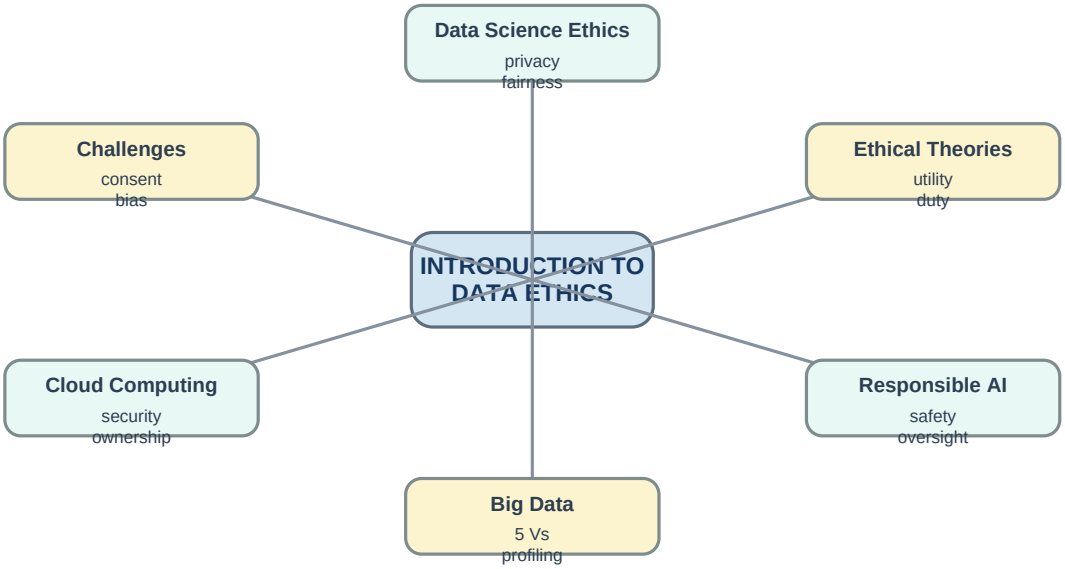
The 5 Vs	Meaning
Variety	Different forms and sources of data
Veracity	Accuracy and reliability of data
Value	Useful insights obtained from data

Big Data ethical concerns include privacy violations, excessive collection, biased datasets, unclear ownership, profiling, security risks, and using data for purposes beyond the original reason.

Cloud Computing is the delivery of services such as servers, storage, databases, networking, and software through the internet. Examples include Google Drive, Microsoft Azure, and Amazon Web Services.

Cloud issue	Explanation
Privacy	Personal information could be exposed.
Security	Cloud systems may face cyberattacks.
Data ownership	Control over stored information may be unclear.
Data location	Data can be stored in different countries and legal jurisdictions.
Unauthorized access	Improper access by employees or attackers can occur.
Vendor dependency	Organizations may become dependent on a provider.
Data loss	Failures or attacks may cause loss if backups are inadequate.

Unit I Mind Map



UNIT II – PRIVACY LAWS AND REGULATORY FRAMEWORKS

1. Overview of Data Protection Laws: GDPR, CCPA, and India

Data Protection Laws are legal rules that govern how personal information is collected, processed, stored, shared, and used. They provide rights to individuals and duties to organizations.

A. GDPR – General Data Protection Regulation

GDPR is the European Union's data protection regulation. It became applicable on 25 May 2018 and aims to protect personal data, support lawful processing, and make organizations accountable.

• **Principles:** lawfulness, fairness, transparency, purpose limitation, data minimization, accuracy, storage limitation, integrity/confidentiality, and accountability.

• **Rights:** access, rectification, erasure under conditions, data portability, restriction, objection, and protections concerning certain automated decisions.

• **Example:** an individual may request a copy of personal data held by an organization.

B. CCPA – California Consumer Privacy Act

CCPA is a California privacy law that gives consumers rights over personal information collected by covered businesses. It became effective in 2020 and was amended by the California Privacy Rights Act (CPRA), with major amendments effective in 2023.

• Rights include knowing what information is collected, requesting deletion or correction, opting out of sale or sharing, limiting certain uses of sensitive information, and non-discrimination.

• Example: a consumer can ask a covered business to disclose categories of personal information it has collected.

C. India: PDPB and DPDP Act

The Personal Data Protection Bill (PDPB) was a proposed Indian law and was withdrawn in 2022. India later enacted the Digital Personal Data Protection Act, 2023 (DPDP Act). The DPDP Act establishes a framework for processing digital personal data, including obligations for Data Fiduciaries and rights for Data Principals. Specific implementation depends on the Act and applicable rules.

Feature	GDPR	CCPA/CPRA	India – DPDP Act 2023
Region	European Union	California, USA	India
Primary focus	Personal data protection	Consumer privacy rights	Digital personal data protection
Consent	One lawful basis among others	Notice and opt-out rights are central in relevant contexts	Consent and certain legally recognized uses
Access rights	Right to access	Right to know	Right to access specified information
Deletion	Right to erasure under conditions	Right to delete under conditions	Right to erasure subject to conditions
Accountability	Organizations must demonstrate compliance	Covered businesses have duties	Data Fiduciaries have duties

2. Right to Privacy: Legal and Philosophical Perspectives

Right to Privacy is the right of individuals to protect personal information, communications, activities, and private life from unnecessary or unlawful interference.

Legal perspective: privacy is protected through laws and constitutional provisions. In India, the Supreme Court recognized privacy as a fundamental right in the 2017 K.S. Puttaswamy judgment. Privacy is protected as part of Article 21 and other constitutional freedoms. Legal protection can provide remedies and constrain unlawful surveillance or data use.

Philosophical perspective: privacy is important because it supports human dignity, autonomy, individual freedom, trust, and the development of personal identity. It allows people to make decisions without unnecessary interference.

Legal perspective	Philosophical perspective
Based on laws and constitutional principles	Based on moral reasoning and values
Provides formal rights and remedies	Explains why privacy is valuable
Enforced through institutions and courts	Influences social and ethical expectations
Example: legal right to access data	Example: respect for personal space

3. Data Ownership and User Consent Mechanisms

Data Ownership refers broadly to rights and responsibilities related to controlling, accessing, managing, and using data. Digital data ownership can be complex: personal data is not always owned like physical property, while privacy laws may provide rights to access, correct, erase, or control processing.

User Consent is permission from an individual for a data practice when consent is the applicable legal basis. Meaningful consent should be freely given, specific, informed, clear, and withdrawable where required.

Consent type	Explanation
Explicit consent	A clear and direct agreement.
Informed consent	Permission after the person understands the purpose and relevant information.
Opt-in	The person actively agrees.
Opt-out	The person can refuse or stop a covered data use.

A responsible consent process explains what data is collected, why it is needed, how it is used, and how the person can exercise choices. Example: a mobile app explains why location access is needed and lets the user decline if the service can operate without it.

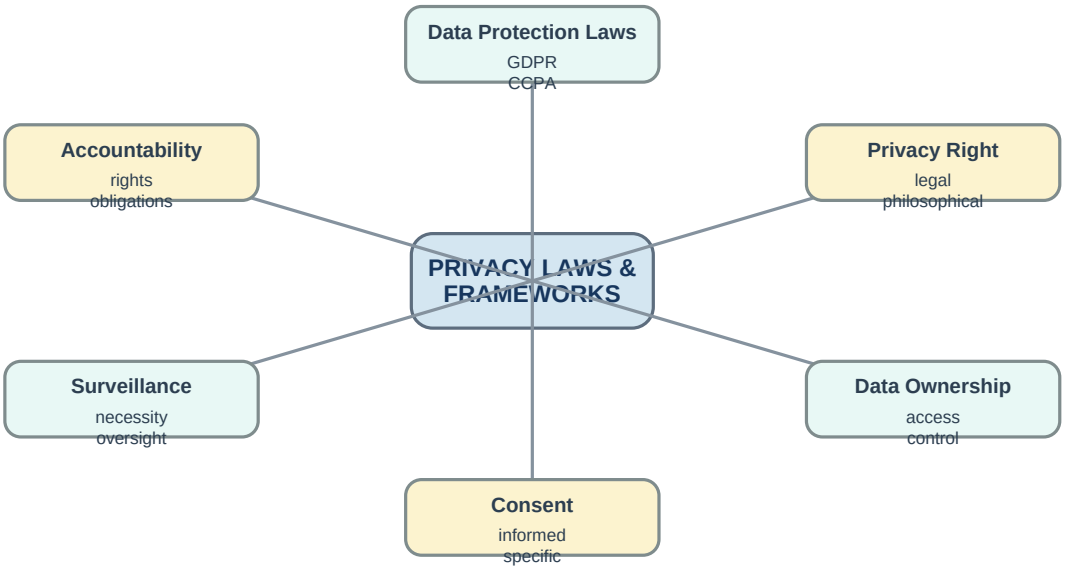
4. Ethical Considerations in Government Surveillance and Data Collection

Government Surveillance is monitoring people, communications, or activities by government agencies for purposes such as national security, crime prevention, and public safety. Examples include CCTV, communication monitoring, and digital activity monitoring.

Potential benefits: crime prevention, national security, law enforcement, and public safety. **Ethical risks:** privacy violations, lack of consent or notice, mass surveillance, misuse of data, discrimination, and limited transparency.

Ethical surveillance requires a lawful basis, legitimate purpose, necessity, proportionality, data minimization, strong security, independent oversight, and accountability. Surveillance should not be broader than necessary for its purpose.

Unit II Mind Map


Data Protection Laws
GDPR
CCPA
Accountability
rights
obligations
Privacy Right
legal
philosophical
PRIVACY LAWS & FRAMEWORKS
Surveillance
necessity
oversight
Data Ownership
access
control
Consent
informed
specific

UNIT III – ETHICAL CHALLENGES IN AI

1. Bias and Fairness in Machine Learning Algorithms

Algorithmic Bias occurs when an AI or machine learning system produces systematic errors or unfair outcomes because of data, design, measurement, or decision-making processes. Models learn patterns from data; if the data reflects historical discrimination, a model may reproduce it.

Cause	Explanation	Example
Biased training data	Dataset contains unfair patterns	Past hiring records favor men
Historical bias	Past social discrimination appears in records	Unequal access represented in outcomes
Sampling bias	Some groups are underrepresented	Face dataset lacks diverse skin tones
Measurement bias	A variable is measured inaccurately	Income data recorded incorrectly
Human bias	Developer assumptions affect design	Unfair feature selection
Labeling bias	Labels are assigned inconsistently or unfairly	Subjective labels in training data

Fairness in ML means providing equitable treatment and avoiding unjust discrimination. Fairness does not always mean identical treatment; context may require different measures to achieve equitable outcomes.

Methods to reduce bias: use representative datasets, correct data errors, evaluate model performance across groups, conduct fairness audits, include diverse teams, document limitations, monitor after deployment, and allow human review for high-impact decisions.

Example: a recruitment model trained on past hiring records selects fewer women because the historical data is male-dominated. A responsible solution includes representative training data, fairness testing, and human oversight.

2. Algorithmic Accountability and Explainability

Algorithmic Accountability means that individuals and organizations responsible for algorithmic systems must be answerable for their decisions and consequences. Accountability involves governance, documentation, audits, monitoring, and channels for affected people to raise concerns.

• **Auditing:** examine algorithms for bias, errors, and risk.

• **Monitoring:** track performance after deployment.

• **Documentation:** maintain records of datasets, design choices, and limitations.

• **Human oversight:** enable review of significant decisions.

• **Complaint mechanisms:** allow people to challenge harmful outcomes.

• **Responsibility assignment:** clarify which teams or organizations are answerable.

Explainable AI (XAI) is a set of methods that makes AI predictions or decisions understandable to people. It can explain which factors influenced a result and help users or professionals identify errors.

Example: a bank's AI rejects a loan request. An explanation may identify factors such as insufficient credit history or high debt. Explanations should be meaningful and not misleading.

Accountability	Explainability
Focuses on responsibility and answerability	Focuses on understanding how a decision was produced
Uses audits, monitoring, governance	Uses explanations and model interpretation
Question: who is responsible?	Question: why did this decision happen?

3. Ethical Implications of Facial Recognition and Biometric Data

Facial Recognition is an AI technology that identifies or verifies people by analyzing facial features and comparing them with images or templates. **Biometric data** is information derived from physical or behavioral characteristics such

as fingerprints, facial features, iris patterns, voice, or hand geometry.

Applications: airport security, phone unlocking, attendance systems, identity verification, access control, and some law-enforcement applications.

Ethical concerns: identification without knowledge, surveillance, unequal accuracy across demographic groups, misidentification, biometric data theft, lack of consent, and function creep (using data for purposes beyond the original purpose).

Ethical safeguards: use accurate and tested systems, obtain consent where required, protect biometric data, limit purposes and retention, avoid discriminatory practices, provide human oversight, and follow applicable laws.

Advantages	Disadvantages
Fast identification	Privacy and surveillance risks
Can strengthen access security	Misidentification may harm innocent people
Can reduce manual work	Accuracy may vary between groups
Convenient authentication	Biometric traits are difficult to replace if compromised

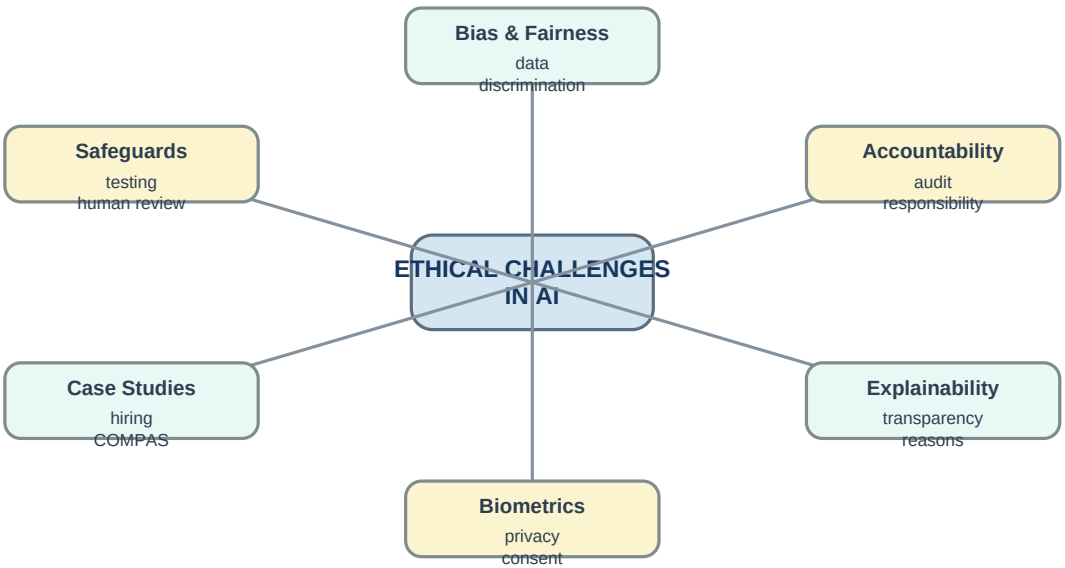
4. Case Studies: AI Ethics Failures and Controversies

Case 1 – Amazon experimental AI hiring tool: Amazon developed an experimental recruitment tool trained on historical resumes. Historical patterns reflected a male-dominated workforce, and the system learned patterns that disadvantaged women in some cases. Ethical concerns included gender bias, unfair recruitment, and biased training data. Lesson: representative datasets, fairness testing, and human review are essential.

Case 2 – COMPAS risk assessment: COMPAS is used in the US criminal justice context to estimate recidivism risk. A 2016 ProPublica investigation reported racial disparities in error rates, including false-positive predictions. Ethical concerns included fairness, transparency, and impacts on rights. Lesson: high-stakes AI requires careful audits, contextual evaluation, and human oversight; fairness metrics can sometimes conflict.

Case 3 – Facial recognition misidentification: Facial recognition systems have wrongly identified people in law-enforcement settings. This can lead to wrongful suspicion, privacy violations, and accountability concerns. Lesson: the technology should not be treated as infallible evidence; human verification and safeguards are necessary.

Unit III Mind Map



UNIT IV – DATA PROTECTION

1. Introduction to Cybersecurity and Ethical Hacking

Cybersecurity is the practice of protecting computers, networks, applications, and data from unauthorized access, attacks, damage, and theft. It aims to preserve confidentiality, integrity, and availability.

CIA principle	Meaning	Example
Confidentiality	Only authorized people can access data	Password and access controls
Integrity	Data remains accurate and unaltered without authorization	Hash verification
Availability	Systems and data are accessible when needed	Reliable backups and servers

Common Cybersecurity Threats

Threat	Explanation
Malware	Malicious software that harms or accesses systems
Phishing	Fake messages designed to steal information
Ransomware	Malware that locks or encrypts data and demands payment
SQL injection	Malicious input used to manipulate database queries
DDoS	Traffic flood that makes a service unavailable
Insider threat	Misuse of access by an authorized person

Ethical Hacking is authorized security testing to identify vulnerabilities in systems, networks, or applications. Ethical hackers must have permission, stay within the agreed scope, protect discovered information, and report weaknesses responsibly.

Hacker type	Description
White Hat	Authorized ethical hacker
Black Hat	Malicious, unauthorized hacker
Grey Hat	May access without authorization, though not necessarily with malicious intent

Typical authorized testing phases: planning and permission → information gathering → vulnerability assessment → scoped testing → report findings → remediation → retesting.

Importance: identifies weaknesses, improves security, prevents breaches, protects confidential data, and builds trust.

2. Encryption Techniques and Secure Data Transmission

Encryption converts readable plaintext into unreadable ciphertext using an algorithm and key. **Decryption** converts ciphertext back into readable plaintext.

Symmetric encryption uses one shared secret key for encryption and decryption. It is fast and suitable for large volumes of data, but the key must be shared securely. Example: AES.

Asymmetric encryption uses a mathematically related public and private key pair. It supports key exchange and digital signatures, but is generally slower for bulk data. Examples: RSA and ECC.

Symmetric	Asymmetric
One shared secret key	Public and private key pair
Fast and efficient for bulk data	Generally slower
Secure key distribution is a challenge	Public key can be shared; private key is kept secret
Example: AES	Examples: RSA, ECC

Secure Data Transmission means transferring data while protecting it against unauthorized reading, modification, or interception. Methods include HTTPS, TLS, digital certificates, encryption, authentication, and VPNs. For example, TLS protects communications between a browser and an HTTPS website.

3. Data Anonymization and Privacy-Preserving Data Sharing

Data Anonymization removes or transforms identifiers so individuals cannot reasonably be identified from the resulting data. Removing names alone may not be enough, because combinations of other attributes can re-identify people.

Technique	Explanation	Example
Masking	Hide sensitive portions	Phone shown as XXXXXXX210
Generalization	Replace precise values with broader groups	Age 21 → age range 20–25
Suppression	Remove sensitive fields	Remove names and phone numbers
Aggregation	Combine records into group statistics	Class average marks
Pseudonymization	Replace identifiers with pseudonyms	Name → Student A
Differential privacy	Add carefully designed statistical noise	Publish aggregate counts with privacy protection

Anonymization vs pseudonymization: effective anonymization aims to prevent reasonable identification; pseudonymization replaces identifiers but may be reversible with additional information and can still be personal data.

Privacy-preserving data sharing shares useful data while limiting privacy risks. Methods include anonymization, encryption, differential privacy, secure multiparty computation, federated learning, and access controls.

Federated Learning trains models across devices or organizations without centrally collecting all raw data. It can reduce exposure, though model updates and outputs still require privacy and security safeguards.

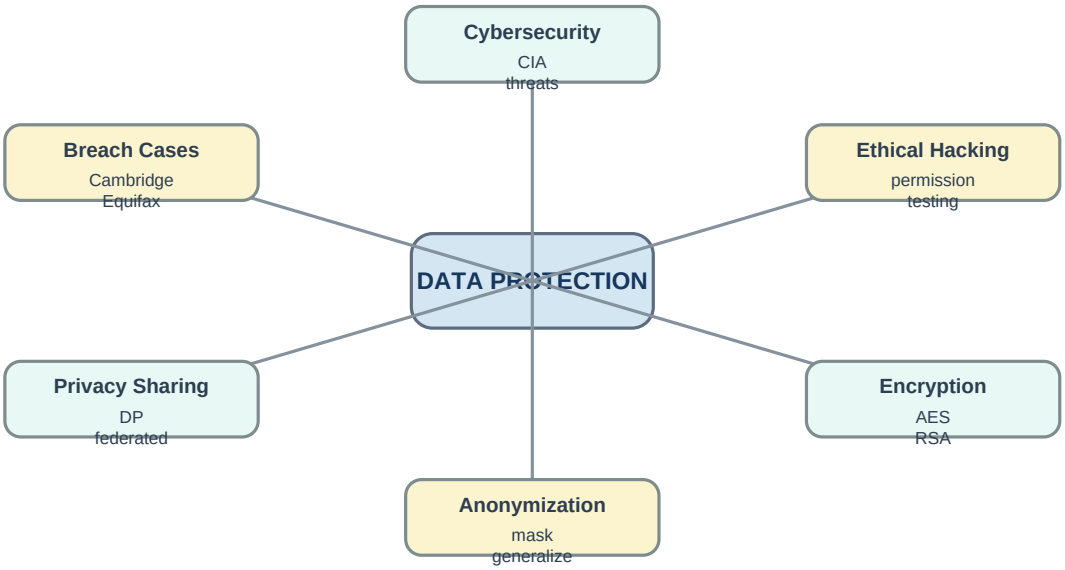
4. Case Studies: Data Breaches and Ethical Implications

Facebook–Cambridge Analytica: data associated with Facebook users was collected through a third-party app and used for political profiling and targeting. Ethical issues included meaningful consent, privacy, transparency, and improper use. Lesson: organizations must limit data use and protect users.

Equifax breach (2017): a major breach exposed personal information of approximately 147 million people. Ethical concerns included security weaknesses, failure to protect sensitive data, identity-theft risks, and risk management. Lesson: patch vulnerabilities, maintain security controls, and respond transparently.

Aadhaar-related privacy concerns: identity and biometric systems raise questions about access, data security, privacy, purpose limitation, and possible misuse. Lesson: sensitive identity data needs strong safeguards, legal protections, and careful access management.

Unit IV Mind Map


DATA PROTECTION
Cybersecurity
CIA
threats
Breach Cases
Cambridge
Equifax
Ethical Hacking
permission
testing
Privacy Sharing
DP
federated
Encryption
AES
RSA
Anonymization
mask
generalize

UNIT V – FUTURE TRENDS IN DATA ETHICS AND GOVERNANCE

1. Emerging Technologies and Their Ethical Implications

Emerging Technologies are new or rapidly developing technologies that may transform society, industries, and everyday life. Examples include AI, IoT, blockchain, quantum computing, robotics, autonomous vehicles, and generative AI.

Technology	Ethical implications	Example concern
AI	Bias, accountability, privacy	Unfair automated decisions
IoT	Privacy, security, consent	Smart devices continuously collect data
Blockchain	Privacy, permanence, energy use	Personal information is hard to erase
Quantum computing	Future cryptographic risks, access inequality	Potential threat to some current encryption
Robotics	Job displacement, safety	Automation affects workers
Generative AI	Misinformation, copyright, bias	Synthetic fake images or text
Autonomous vehicles	Safety and accountability	Responsibility after a crash

Ethics is necessary to protect rights, promote responsible innovation, reduce harm, support public safety, preserve privacy, and build trust.

2. Ethics in IoT, Blockchain, and Quantum Computing

A. Internet of Things (IoT)

IoT is a network of physical devices connected to the internet that collect and exchange data. Examples include smart watches, smart home devices, connected vehicles, and health monitors.

Ethical issues include privacy (continuous data collection), security (hacking), consent (users may not understand collection), surveillance, data ownership, and physical safety. Solutions include strong passwords, encryption, data minimization, clear notices, consent where required, and regular updates.

B. Blockchain

Blockchain is a distributed digital ledger that records transactions in linked blocks using cryptographic techniques. Common characteristics include decentralization, transparency, immutability, security, and traceability.

Ethical issues include public exposure of transaction details, difficulty erasing permanent records, energy use of some consensus mechanisms, unclear accountability, and potential misuse. Ethical approaches include not storing sensitive personal data directly on-chain, privacy-preserving design, energy-efficient consensus, governance, and legal compliance.

C. Quantum Computing

Quantum computing uses quantum-mechanical principles and qubits to process information. Potential applications include scientific research, optimization, drug discovery, material development, and cryptography research.

Ethical issues include future threats to some existing cryptography, privacy risks for stored data, unequal access, harmful misuse, and governance challenges. Solutions include preparing quantum-resistant cryptography, responsible research, security standards, fair access, and international cooperation. Current quantum computers are not generally capable of breaking modern cryptography at large scale; the concern is future capability and long-term protection.

3. Corporate Social Responsibility in Data Management

Corporate Social Responsibility (CSR) in data management means that organizations have a responsibility to collect, process, store, and use data ethically, securely, transparently, and in ways that consider society and the environment. A company should not focus only on profit; it should also protect users and communities.

Responsibility	Explanation
Data privacy	Protect personal information
Data security	Prevent unauthorized access and attacks
Transparency	Explain how data is collected and used
Fairness	Avoid biased decisions and discriminatory practices
Accountability	Accept responsibility for misuse or failures
Sustainability	Reduce environmental impact of computing and data centers
Legal compliance	Follow applicable laws and regulations

CSR benefits include customer trust, improved reputation, protection of individuals, fewer legal risks, ethical business practices, and sustainable development. Example: a company encrypts customer data, limits access, provides clear notices, and avoids unauthorized sharing.

4. Developing an Ethical Code of Conduct for Data Professionals

Ethical Code of Conduct is a set of rules and principles that guides data professionals—including data scientists, AI developers, database administrators, and cybersecurity professionals—to behave responsibly.

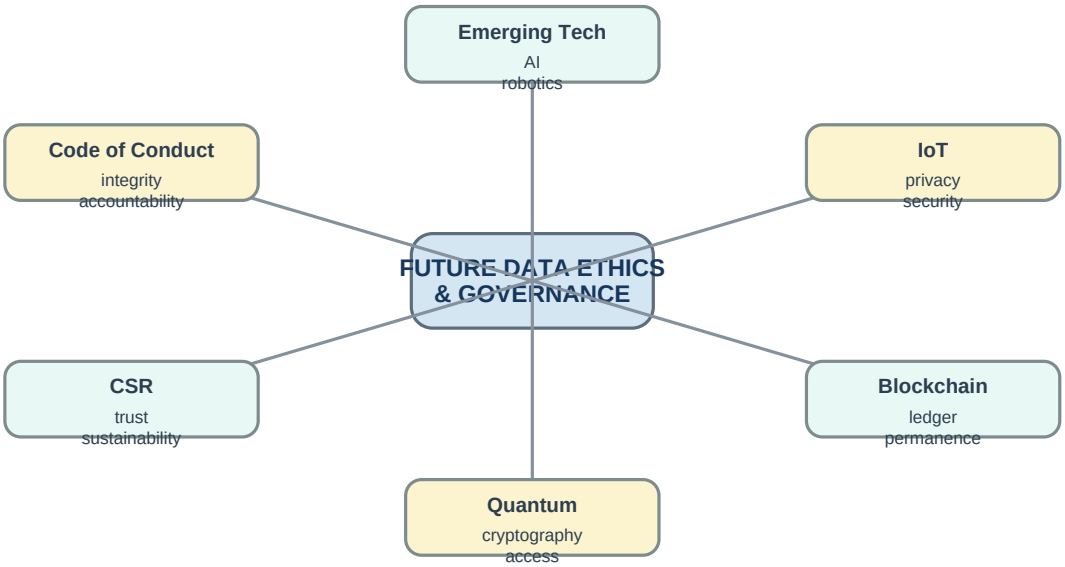
Principle	Expected conduct
Privacy	Protect personal and sensitive information
Honesty	Report data and results accurately
Fairness	Avoid discriminatory algorithms
Confidentiality	Keep sensitive information protected
Transparency	Explain data practices appropriately
Accountability	Accept responsibility for decisions
Security	Protect data from threats
Competence	Maintain technical skills and knowledge
Consent	Respect permission and choices
Integrity	Do not manipulate or falsify data

Steps to develop the code: (1) identify ethical issues, (2) define principles, (3) write rules and guidelines, (4) assign responsibilities, (5) train professionals, (6) implement the code, (7) monitor and audit, and (8) review and update regularly.

Example conduct: collect data for legitimate purposes; obtain consent when required; protect confidential information; avoid biased models; report errors honestly; never misuse personal data; follow laws; and accept responsibility for mistakes.

Benefits include ethical behavior, reduced misuse, public trust, professionalism, responsible data management, and organizational accountability.

Unit V Mind Map


FUTURE DATA ETHICS & GOVERNANCE
Emerging Tech
AI
robotics
Code of Conduct
integrity
accountability
IoT
privacy
security
CSR
trust
sustainability
Blockchain
ledger
permanence
Quantum
cryptography
access

QUICK REVISION

Unit	Core topics
Unit I	Data Ethics, ethical theories, Responsible AI, Big Data, Cloud Computing
Unit II	GDPR, CCPA, PDPB, DPDP Act, privacy rights, consent, surveillance
Unit III	Bias, fairness, accountability, XAI, facial recognition, AI case studies
Unit IV	Cybersecurity, ethical hacking, CIA Triad, encryption, anonymization, breaches
Unit V	IoT, Blockchain, Quantum Computing, CSR, Ethical Code of Conduct

Important differences to revise

Pair	Key distinction
Utilitarianism vs Deontology	Consequences vs duties/rules
Deontology vs Virtue Ethics	Rules/duty vs character
GDPR vs CCPA	EU data protection regulation vs California consumer privacy law
Bias vs Fairness	Systematic skew or unfairness vs equitable treatment
Accountability vs Explainability	Responsibility vs understanding the decision
Symmetric vs Asymmetric encryption	One shared key vs public/private key pair
Encryption vs Anonymization	Make data unreadable with a key vs reduce identifiability
Anonymization vs Pseudonymization	Aim to prevent reasonable identification vs replace identifiers but retain potential linkability

Important 10-Mark Questions

• Explain the principles and challenges of Data Ethics.

• Discuss Utilitarianism, Deontology, and Virtue Ethics.

• Explain Responsible AI and ethics in automation.

• Explain GDPR, CCPA, and India's data protection framework.

• Discuss the Right to Privacy from legal and philosophical perspectives.

• Explain bias and fairness in machine learning algorithms.

• Discuss algorithmic accountability and Explainable AI.

• Explain ethical implications of facial recognition and biometric data.

• Explain cybersecurity, ethical hacking, and the CIA Triad.

• Discuss symmetric and asymmetric encryption techniques.

• Explain data anonymization and privacy-preserving data sharing.

• Discuss data breach case studies and their ethical implications.

• Explain ethics in IoT, Blockchain, and Quantum Computing.

• Discuss CSR in data management.

• Explain the steps to develop an Ethical Code of Conduct.

Exam answer structure

For 5 marks: definition, 4–5 key points, and one example. **For 10 marks:** introduction/definition, organized headings, detailed points, table or diagram when useful, example, and short conclusion.

END OF NOTES